

Suhyun Choi

010-8912-1828 | chsuk02@hanyang.ac.kr

RESEARCH INTERESTS

3D Vision, 6D Object Pose Estimation, Robot Calibration, and Visual Grounding

EDUCATION

Hanyang University (ERICA)

Ansan, South Korea

B.S. in Robot Engineering (GPA: 3.62 / 4.50 | Major GPA: 3.97 / 4.50)

Mar 2021 - Present

EXPERIENCE

Undergraduate Research Assistant

Mar. 2026 – Apr. 2026

RoCogMan Lab, Hanyang University ERICA

Ansan, South Korea

- Implemented calibration and 3D reconstruction pipeline for a stage-based line-laser scanning system

Undergraduate Research Assistant

Jan. 2026 – Feb. 2026

Korea Electrotechnology Research Institute (KERI)

Ansan, South Korea

- Assisted research on medical image processing to estimate tissue oxygen saturation (StO2) using GANs with hyperspectral and RGB cameras

RESEARCH OUTPUT

S. Choi et al., **Deep Learning-Based Tissue Oxygen Saturation Map Estimation Using RGB Images**

Poster Presentation (Accepted), KOSOMBE 2026 Spring Conference

PROJECTS

Physical Security Mobile Robot System | *Capstone Design with Kakao Data Center*

Sep 2025 – Present

- Built autonomous patrol robot for data center security
- Developed feature-based scene change detection and YOLO-based human tracking
- Implemented FastAPI server for robot control, monitoring, and event logs

RB5 Manipulator Vision-Based Pick-and-Place

Mar 2026 – Jun 2026

- Performed hand-eye calibration using RGB-D camera
- Implemented 6D pose estimation for robotic pick-and-place

AI Autonomous Driving Contest

Sep 2025 – Nov 2025

- Developed lane detection and YOLO-based obstacle recognition
- Integrated ROS Ackermann steering control

Reverse Distillation-Based Anomaly Detection

Jan 2025 – Feb 2025

- Implemented reverse distillation-based anomaly detection for industrial defect inspection
- Replaced ResNet backbone with ConvNeXt to improve feature representation

TECHNICAL SKILLS

Python, C, ROS2, PyTorch, OpenCV, Open3D, NumPy, Git, Ubuntu Linux

English: TOEIC 873

Certifications: KT AICE Associate

LEADERSHIP & SERVICE

Executive Officer *HY-MEC Robotics Engineering Society*

Mar 2025 – Dec 2025

- Organized robotics hackathon and technical seminars

Squad Leader & Drill Instructor *Korea Army Training Center, Nonsan*

Apr 2023 – Oct 2024

- Led recruit training and daily unit operations